

# Robert Keller

Senior Data Engineer

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## SUMMARY

Senior Data Engineer with **10+ years** designing large-scale, cloud-native data platforms across **fintech**, **healthcare**, and **e-commerce**. Expert in building real-time streaming systems using Kafka, Spark Structured Streaming, Delta Lake, and modern **lakehouse** architectures on **AWS**, **Azure**, and **Databricks**. Proven track record modernizing legacy ecosystems into high-throughput, governed platforms using **Airflow**, **dbt**, **Snowflake**, **Redshift**, and secure frameworks such as **IAM**, **Lake Formation**, **Purview**, and **KMS**. Adept at partnering with ML teams to deliver production-grade feature pipelines and driving enterprise-wide data quality, observability, and CI/CD automation using **Python**, **Terraform**, and **Docker**.

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## TECHNICAL SKILLS

- **Programming:** Python, Scala, Java, Bash
  - **Data Processing & ETL:** Apache Spark, Apache Airflow, dbt, Apache Beam
  - **Data Warehousing:** Snowflake, Amazon Redshift
  - **Cloud Platforms:** AWS (S3, Glue, EMR, Lambda, Athena, Redshift, Kinesis), Azure (Data Factory, Synapse, Blob Storage, Event Hubs)
  - **Data Lakes & Storage:** Amazon S3, Azure Data Lake Storage, Delta Lake, Iceberg
  - **Streaming & Messaging:** Apache Kafka, Apache Flink
  - **DevOps & Infrastructure:** Docker, Kubernetes, Terraform, CI/CD
  - **Monitoring:** Prometheus, Grafana, Datadog, ELK Stack
  - **Data Governance & Quality:** Great Expectations, Apache Atlas, Data Contracts
  - **BI & Visualization:** Tableau, Looker, Power BI
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## PROFESSIONAL EXPERIENCE

**GDIT – Sr Data Engineer | Falls Church, VA | 07/2022 – Present**

- Architected a unified real-time data backbone for a **fintech platform**, integrating payments and order events using **Kafka**, **Spark Structured Streaming**, and **AWS MSK**, enabling millisecond-level fraud scoring, checkout anomaly detection, and seamless transaction monitoring across the system.
- Engineered a multi-zone **Delta Lake** platform on **AWS S3** with ACID guarantees, powering scalable analytics for transaction, cart, and fulfillment datasets across multiple business units.
- Modernized 180+ legacy ETL jobs into **Airflow** and **dbt**, improving pipeline modularity and reducing SLA breaches by **45%** during peak shopping periods.
- Built a metadata-driven ingestion engine using **Python**, **Jinja2**, and **YAML**, accelerating onboarding of new payment gateways, loyalty systems, and vendor feeds.

- Designed cross-domain data contracts and schema evolution workflows to ensure consistency between streaming fraud models, batch settlement pipelines, and **e-commerce** catalog systems.
- Integrated **Great Expectations** and **OpenMetadata** to automate profiling, lineage, and validation for high-value datasets including transactions, refunds, and chargebacks.
- Designed and orchestrated large-scale batch and streaming workflows in **Apache Airflow**, coordinating cross-domain pipelines for payments, orders, fraud scoring, and customer analytics.
- Delivered end-to-end ingestion pipelines on **Databricks** using **Auto Loader** and **Delta Live Tables**, enabling scalable processing of product, inventory, and behavioral event streams.
- Tuned **Snowflake** performance through clustering, micro-partition pruning, and warehouse right-sizing, reducing compute spend by **45%** across analytics workloads.
- Implemented a secure, multi-tenant data architecture using **AWS IAM**, **Lake Formation**, and **KMS** to meet **PCI-DSS**, **SOC 2**, and **e-commerce** fraud-prevention compliance requirements.
- Built streaming feature pipelines for credit-risk scoring, fraud detection, and personalized product recommendations in partnership with **ML engineering teams**.
- Developed CI/CD automation for data infrastructure using **GitHub Actions**, **Terraform**, and **Docker**, enabling consistent deployments across dev, staging, and production.
- Implemented **Redshift Spectrum** and federated queries to blend **S3-based data lake** assets with warehouse tables, reducing duplication and improving data freshness.
- Led architecture reviews and mentored engineers on cloud-native design patterns, high-throughput streaming, and secure data sharing across **fintech** and **e-commerce** domains.
- Delivered real-time dashboards in **Looker** and **Power BI** for fraud alerts, order velocity, and revenue KPIs, improving visibility for risk, finance, and operations teams.

### ***Tenet Healthcare – Data Engineer | Dallas, TX | 04/2019 – 06/2022***

- Engineered a full-cycle **healthcare** data modernization initiative, migrating legacy **ETL** workflows and analytics systems into a real-time, cloud-native **Azure data platform**.
- Developed scalable **ETL/ELT pipelines** using **PySpark**, **SQL**, and **Azure Data Factory**, enabling automated processing of clinical, claims, and EHR datasets across multiple business units.
- Designed a multi-zone **Azure Data Lake Storage (ADLS)** architecture with raw, staged, and curated layers, implementing lifecycle policies, versioning, and governed retention.
- Built streaming ingestion pipelines using **Kafka** and **Spark Structured Streaming** to capture patient vitals and device telemetry, reducing data latency from 24 hours to under 30 minutes.
- Modeled patient-centric analytics tables in **Snowflake**, optimizing clustering and micro-partitioning to improve cohort query performance by **60%**.
- Implemented HIPAA-aligned data protection using **Azure Key Vault**, **Purview**, **RBAC**, and **ADLS ACLs**, ensuring secure handling of PII/PHI across ingestion and consumption layers.
- Designed and orchestrated large-scale **ETL/ELT pipelines** on **Databricks** using **PySpark**, **Delta Lake**, and **workflow** jobs, enabling reliable processing of multi-terabyte healthcare datasets.
- Created a unified enterprise data catalog using **Azure Purview**, enabling lineage tracking, glossary management, and standardized metric definitions across engineering, analytics, and clinical teams.
- Developed **dbt** models to transform raw claims and encounter data into curated, analytics-ready datasets supporting clinical operations and regulatory reporting.
- Built an **Iceberg**-based **lakehouse** layer to support **ACID** transactions, schema evolution, and time-travel queries for high-volume healthcare datasets.

- Integrated **FHIR** and **HL7** standards into ingestion pipelines, enabling interoperability with external providers, payers, and partner systems.
- Automated metadata-driven pipeline generation using **Jinja2** templates and **YAML** configs, reducing onboarding time for new data sources.
- Containerized ingestion and transformation workloads using **Docker** and deployed them on **AKS**, improving reliability and environment consistency.
- Developed end-to-end observability using **Datadog**, **Grafana**, and **Azure Monitor**, unifying infrastructure, application, and pipeline metrics for proactive SLA monitoring.
- Collaborated with data scientists to deliver ML-ready feature datasets for predictive risk models, supporting initiatives in readmission prediction, chronic disease management, and patient stratification

### ***Nutanix – Data Analystist | San Jose, CA | 07/2015 – 03/2019***

- Contributed to the delivery of a multi-tenant **SaaS analytics platform**, shaping data requirements, validation rules, and reporting layers used by internal teams and external customers.
- Designed KPI dashboards and analytical models using **Tableau**, **Looker**, and **SQL**, improving stakeholder visibility into product usage and customer behavior.
- Built automated data preparation workflows in **Python** to clean, normalize, and enrich incoming SaaS event data, reducing manual prep time by **45%**.
- Partnered with engineering to define data contracts and ensure consistent schema evolution across ingestion, transformation, and reporting layers.
- Analyzed product telemetry and user activity logs to identify adoption patterns, influencing roadmap decisions for onboarding and feature rollout.
- Developed SQL-based data marts supporting finance, marketing, and customer success, enabling faster self-service reporting across teams.
- Created reproducible data quality checks using **Python** and **SQL** to validate incoming datasets and flag anomalies before they reached production dashboards.
- Collaborated with data engineers to map business metrics to underlying data sources, ensuring accuracy across the SaaS platform's analytics layer.
- Conducted deep-dive analyses on churn, retention, and cohort behavior, uncovering insights that contributed to a **45%** improvement in targeted upsell campaigns.
- Documented data lineage, metric definitions, and dashboard logic to support onboarding and reduce dependency on engineering teams.
- Worked with product managers to translate business questions into analytical requirements and deliver actionable insights for feature prioritization.
- Supported **A/B testing** and **experiment analysis**, providing statistical summaries and visualizations to guide product decisions.
- Assisted in validating new data pipelines and integrations during platform releases, ensuring analytics continuity across environments.

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## **EDUCATION**

### ***University of Florida | Gainesville, FL | 09/2011 – 06/2015***

BS, Computer Science